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METHANATION – ROUND 1

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Abstract

The methanation of carbon dioxide (CO_2) is a valuable tool for reducing the emission of greenhouse gases into the environment while injecting synthetic natural gas into the gas grid as a renewable resource. In this set of experiments, we describe the use of CO_2 and a 10% wt. % Ni/Al_3O_2 catalyst of unknown composition within a continuously stirred tank reactor. The evaluation of reactor volume, influence of contact time, and influence of temperature, all at stoichiometric ratios were studied. The reactor was found to have a volume of approximately 305 mL. Additionally, the reaction constant was experimentally determined to be 0.301 ± 0.012, 0.4741 ± 0.009, 0.6293 ± 0.019 at 250, 270, and 300 °C. Derived from this data, the apparent on energies were found to be 5.81 and 37.57 kJ/mol while the frequency factors were found to be 28.28 and 2.40 s^{-1}. Contract time was found to have a positive relationship with the conversion of CO_2. Understanding the reaction parameters for methanation allows for the control of the reaction, potentially allowing for the large-scale reduction of CO_2 emissions while contributing to the natural gas network.

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Introduction

It is no secret that greenhouse gas emissions are at an all-time high, with CO_2 levels reaching unseen record levels due to the exploitation of fossil fuels. According to the World Meteorological Association (WMA) there are over 40 million tons of CO_2 released per year[1], which drives global warming to dangerous levels. Due to this, environmental awareness has played an increasingly key role in everyday life, and so, many solutions to convert released CO_2 into alternative fuel sources. One of these solutions is through the conversion of CO_2 with H_2 – especially the H_2 produced through electrolysis[2] – into methane, this process is called methanation, whose most common form is through the Sabatier reaction, reported in eqn. 1[2]:

$$CO_2 + 4H_2 \leftrightarrow CH_4 + 2H_2O \qquad \Delta H_{298K} = -165\ kJ/mol \quad (1)$$

The process of this type of methanation was initially proposed by Paul Sabatier and Jean-Baptiste Senderens in 1902 with the idea in mind to remove CO_x within the ammonia synthesis process[3]. Beyond emissions reduction, methanation has been proven to enable power-to-gas (PtG), with allows for storing surplus renewable electricity as readily accessible and dispatchable synthetic methane. Such synthetic methane is compatible with the existing natural gas pipelines and storage units – which in turn could lower the cost of seasonal storage and transport[4]. Industrially, the process of methanation has also been used for other processes such as removal of CO_x that could be harmful to catalysts, thus improving their efficiency. Thus, the methanation process turns waste climate liability into a valuable fuel while supporting energy security.

Our experimental set-up consists of a Horiba® VA-3000 gas analyzer connected to CO_2 and N_2 cylinders and connected to a Continuously Stirred Tank Reactor (CSTR) containing a Ni/Al_2O_3 catalyst. It is important to note that the CO_2 cylinder contains only 4.015 Vol% CO_2 with the rest Vol% consisting of pure N_2. Our purpose is to calibrate the VA-3000 gas analyzer to further analyze the Sabatier reaction – in specific, analyze the reactor volume of the CSTR and then compare it with the manufacturer's specification, the overall kinetic reaction order, understand the relationship between temperature and contact time, estimate the overall activation energy of the reaction, and then compare the results with those found in literature. In order to attain these results, we have determined that some assumptions are to be made: ideal gas, steady state, process is isothermal, uniform surface distribution of the catalyst, and that the reaction follows Langmuir-Hinshelwood kinetics such that the surface dissociation step is the rate determining step.

Background & Theoretical Analysis

Methane synthesis is a kinetics-driven process taking place on a metal catalytic surface. Key reactions happening in the methanation reactor are listed below [5]. Methanation is highly exothermic, driven by primary reactions (a) and (b). By adjusting the ratio CO/H_2, the hydrogenation of CO could yield side products in (c). At lower or intermediate temperatures,

coke (carbon) formation and deposition on the catalytic surfaces can occur. This eventually blocks active sites and reduces synthesis effectiveness.

$$CO_{(g)} + 3H_{2(g)} \leftrightarrow CH_{4(g)} + H_2O_{(g)} \qquad \Delta H_{298K} = -206\ kJ/mol \quad \text{(a)}$$

$$CO_{2(g)} + 4H_{2(g)} \leftrightarrow CH_{4(g)} + 2H_2O_{(g)} \qquad \Delta H_{298K} = -165\ kJ/mol \quad \text{(b)}$$

$$CO_{(g)} + H_2O_{(g)} \leftrightarrow CO_{2(g)} + H_{2(g)} \qquad \Delta H_{298K} = -41\ kJ/mol \quad \text{(c)}$$

$$2CO_{(g)} \leftrightarrow C_{(s)} + CO_{2(g)} \qquad \Delta H_{298K} = -172\ kJ/mol \quad \text{(d)}$$

$$CH_{4(g)} \leftrightarrow C_{(s)} + 2H_{2(g)} \qquad \Delta H_{298K} = -75\ kJ/mol \quad \text{(e)}$$

There are two commonly accepted mechanisms of CH_4 formation: the formate pathway (associative) and the CO pathway (dissociative) as shown in Figure 1[6]. Depending on which type of catalyst, the reaction can undergo either pathway to produce methane.

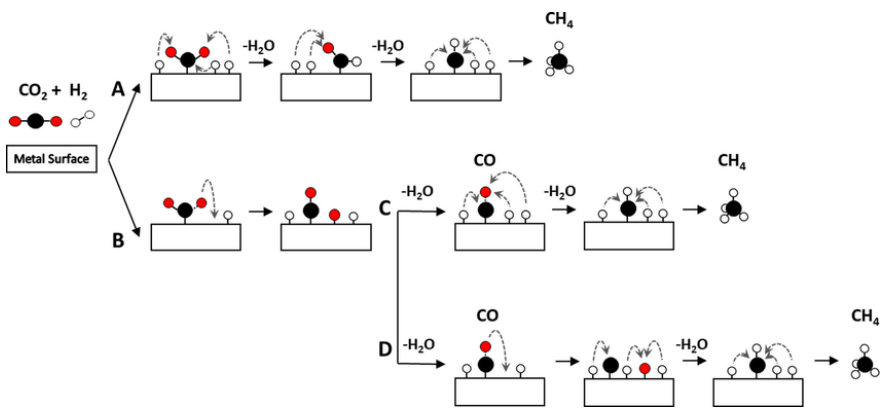

Figure 1: Associative (top) & dissociative (down) pathways for methanation [6].

In this study, Ni/Al was used for the methanation reaction, which was conducted inside a continuous stirred-tank reactor (CSTR), a device that allows well-controlled conditions and residence time to study reaction kinetics. By constant mixing, it maintains uniform concentration and temperature throughout its volume for reactions to occur at steady state. Assuming constant density, volume, uniform surface, and perfect mixing, the reactor volume can be determined from equation 4. At each time point, t, where C_0 is the initial carbon dioxide (Vol %), $\Delta C_{A,in}$ represents the induced step-change for every interval of 10 seconds, and Q is the total flow rate.

$$V = \frac{-qt}{\ln\left(1 - \frac{C_A(t)}{\Delta C_{A,in}}\right)} \qquad (4)$$

The next goal was to experimentally identify the reaction order using the dimensionless variable M_{An} which represents the relationship between the inlet concentration $C_{A,0}$ and outlet/inlet ratio $C_A/C_{A,0}$. For each reaction order, their relationships are described as nonlinear

models shown in Table 1. For each temperature, the model corresponding to a reaction order that best describes $C_{A,0}$ and the experimental outlet/inlet ratio $C_A/C_{A,0}$ is selected and k_A can be computed using equation 5.

Table 1: Relationships between C_A and $C_A/C_{A,0}$[7]

Order (n)	$C_A/C_{A,0}$	$M_{A,n}$
-1st	$\frac{1-\sqrt{1-4(pC_{A,0}^2)}}{2}$	$pC_{A,0}^{-2}$
0th	$1-pC_{A,0}^{-1}$	$pC_{A,0}^{-1}$
½	$\frac{1-(pC_{A,0}^{-1/2})}{2}\left[\left(1+\frac{4}{pC_{A,0}^{-1/2}}\right)^{1/2}-1\right]$	$pC_{A,0}^{-1/2}$
1st	$(1+p)^{-1}$	p
3/2	$pC_{A,0}^{1/2}\left(\frac{C_A}{C_{A,0}}\right)^{3/2}+\frac{C_A}{C_{A,0}}-1$	$pC_{A,0}^{1/2}$
2nd	$\frac{(1+4pC_{A,0})^{1/2}-1}{2(pC_{A,0})}$	$pC_{A,0}$
3rd	$(pC_{A,0}^2)\left(\frac{C_A}{C_{A,0}}\right)^3+\left(\frac{C_A}{C_{A,0}}\right)-1$	$pC_{A,0}^2$
Where $p=k_At$ (5)		

The Arrhenius equation (6) relates temperatures to rate constants, where E_a is the activation energy, R is the gas constant, A is the frequency factor, and T is the absolute temperature. Given variations in temperature and rate constant, the activation energy and the frequency factor can be found using equation 6.

$$\ln(k) = -\frac{Ea}{R}\left(\frac{1}{T}\right) + \ln(A) \qquad (6)$$

The impact of contact time on CO_2 conversion can be studied by relating residence time-calculated from reactor volume and total flow rate to the experimental conversion calculated by:

$$X = 1 - \frac{C_A}{C_{A,o}} \quad (7)$$

The kinetic model for methanation described in literature is reportedly the Langmuir-Hinshelwood (L-H) mechanism. Specifically, for the Ni/Al_2O_3 catalyst, the proposed rate expression for an L-H reaction is below[8]:

$$r_{CH_4} = \frac{k p_{CO_2}^{1/2} p_{H_2}^{1/2}}{\left(1 + K_1 p_{H_2}^{1/2} + K_2 p_{CO_2}^{1/2} + K_3 p_{H_2O}\right)^2} \qquad (8)$$

According to the L-H mechanism, reactants are adsorbed on the surface before reacting with each other. Both adsorbed CO_2 and H_2 compete for active sites, leading to their respective adsorption equilibrium constants, K_1, K_2, K_3, for H_2, CO_2, H_2O, respectively. Followed by the dissociation of CO_2 to CO and O, the rate expression (8) was further developed based on the assumption of CO dissociation being the rate-limiting step, and the abundant species on the catalytic surface being H, CO, and H_2O. The kinetic and equilibrium parameters, including K, were extrapolated to desired temperatures in the study and were determined by least-squares fitting to experimental data. Moreover, the surface coverage of adsorbed reactants is dependent on the partial pressures p_{H_2}, p_{CO_2}, p_{H_2O} which explains the variation in model behavior at different pressures and accounts for any inhibitory effects.

Beyond the intrinsic kinetics of methanation described L-H mechanism, the overall rate is also influenced by the transport phenomena. Under experimental conditions, the process can fall into reaction-limited region, diffusion-limited region, or external mass transfer control. The plot [9] below shows how effective reaction rate, k_{eff} changes with inverse temperature, which highlights the transitions between different operating regime. At higher temperatures, given the low activation energy, the reaction rate is limited by external mass transfer, whereas at lower temperatures, the kinetic control dominates the system.

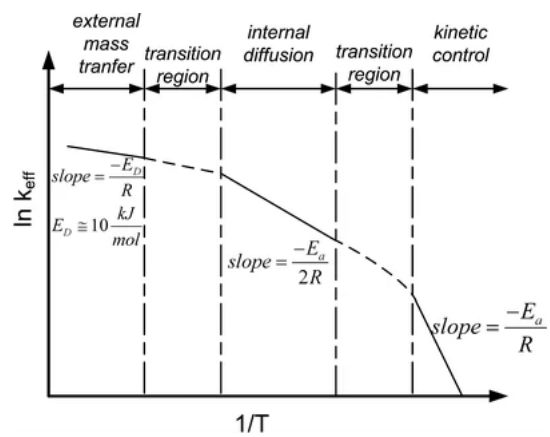

Figure 2: Arrhenius plot illustrating transition from kinetic to mass transfer control [9]

Apparatus & Materials

This experiment was conducted using a 300 mL CSTR manufactured by Parr Instrument [10]. The reactor maintains constant heating with the accuracy of ±1 °C, monitored by a controller also from Parr Instrument, and was held at a constant 200 RPM throughout the course of all experiments. The composition of the gas was controlled using a rotameter, through the software LabVIEW.

Hydrogen gas (18% H_2, 82% N_2) and carbon dioxide gas (4.015% CO_2, 95.985% N_2) were utilized as reactants. Pure N_2 was utilized as an inert gas. The catalyst within the reactor was 10 wt. % Ni/Al_3O_2.

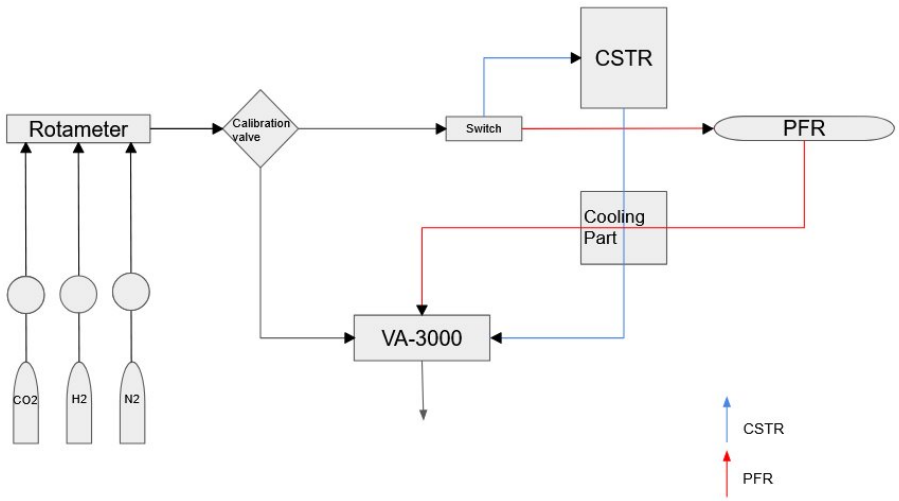

Figure. 2: P&ID of the Methanation Configuration

Experimental Procedure

Calibration

Run 100 SCCM of pure N_2 through the "calibration mode" of the VA-3000 and follow the manufacturer's instructions on how to calibrate the apparatus. Following the tare and span set-up run 100 SCCM of CO_2 along with the N_2, if calibrated properly, a value close to 2 Vol% should be displayed in the IR.

Additionally, pure nitrogen was used to purge the CSTR and VA-3000 Analyzer between every experimental run.

Reactor Volume

Through the CSTR run 2 Vol% of CO_2 and take the time it takes to reach that Vol% in the IR display – note that there is a dead time, only begin timing when the first change is seen in the IR display. Jot the values obtained per ten second intervals and perform an iterative loop by coding (3) while computing the average volume of the reactor, then compare the value to the manufacturer's specifications.

Reaction Order & Rate Constant

To determine the overall kinetic order Θ, run an isothermal and isochoric process while varying N_2, H_2, & CO_2 composition, determine conversion ($C_A/C_{A,0}$). When both C_A & $C_A/C_{A,0}$ have been determined, fit the data to match data according to Table 1, the line of best fit for your data will determine Θ – perform this analysis with various temperatures. Utilizing the parameter $p = k_A\tau$, an experimental reaction constant k can be deduced by letting $\tau = Vq^{-1}$.

Apparent Activation Energy and Frequency Factor

Compose an Arrhenius plot to graphically determine the apparent activation energy E_a and the frequency factor A. Once the values for Θ, k, E_a, and A have been found, compare with known literature.

Impact of Contact Time on Conversion

To evaluate the impact of contact time on conversion, produce a plot of CO_2 conversion vs τ. This was done by maintaining the stream's concentration while varying the total flowrate of the gas flow.

Shutdown

Make sure the heater is off. Shutoff CO_2 flow from the LabView interface, then run only 100 SCCM H_2/N_2 and ensure that the gas cylinders are not empty.

Results & Discussion

Calibration

The calibration of the VA-3000 showed a slight deviation from the calculated inlet rate calculated based off the flowrates from the flowmeter. Calibration curves were created based on the deviation from calculated inlet values (see Appendix section: Calibration).

Reactor Volume:

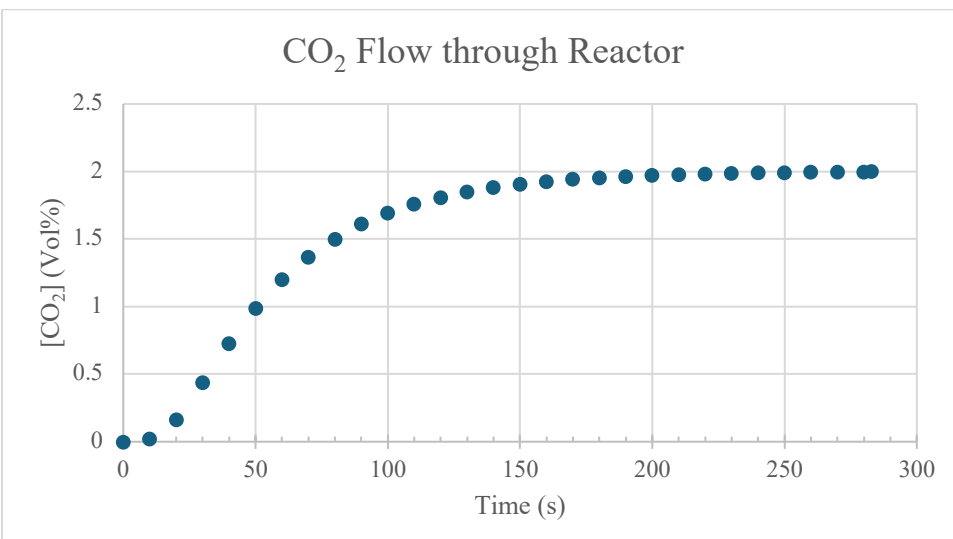

Figure 3: CO_2 Flow through a CSTR

The volume of the CSTR was calculated utilizing the time in which the reactor maintained linearity (dismissing the deadtime) with a step-induced change from 0 to 2 vol % CO_2. From equation (3), the average volume of the reactor was found; it was assumed that the concentration of CO_2 at the beginning of the step change was null. The estimated reactor volume was found to be 305.38 $\pm$ 24.45 mL, consistent with manufacturer specifications (300 mL)[10] .

Inconsistencies may be due to trace amounts of CO or CH_4 remaining in the reactor, as well as the volume between the VA-3000 analyzer and the piping connecting the gas cylinders, the CSTR, and the analyzer. More specifically, the deadtime induced by the piping and analyzer could be a reason as to why the behavior of the residence time differs from that of an ideal CSTR.

Reaction Order & Reaction Contant

To obtain the overall reaction order Θ and the reaction constant k, we varied the compositions of gas stream while maintaining the total flowrate in an isothermal and isochoric process. We ran this method at three different temperatures: 250, 270, and 300 °C.

To analyze the results, we utilized the known relationships between $C_{A,0}$ and $C_A/C_{A,0}$ at various reaction orders – summarized in Table 5 – and created linearized models of each relationship (see Appendix section: MATLAB codes). $C_{A,0}$ and $C_A/C_{A,0}$ collected at various temperatures, summarized in Tables 4, 6, and 7, were also inputted into MATLAB. The method of least squares was used to best fit the data to the pre-established linear models with respect to each reaction order.

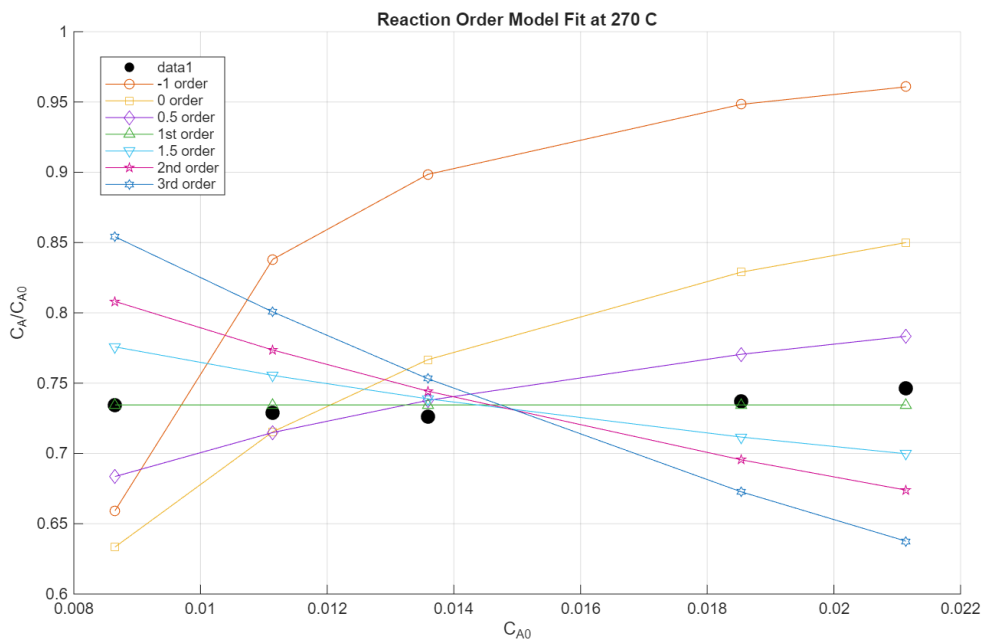

Figure 4: Reaction Order Model at 270 °C

The figures for 250 °C and 300 °C are summarized in the appendix section Computation of Reaction Order.

From Figure 4, it can be observed that the experimental set behaves similarly to the linearized fit for a first order reaction, ascertaining that $\Theta = 1$ within the range of 250 – 300 °C. This behavior is consistent with findings at 250 °C and 300 °C. It can be concluded that the methanation reaction follows first order reaction within the experimental temperature range.

The determination that methanation is first order within our temperature range suggests that Langmuir-Hinshelwood mechanism is a valid assumption for this model. Based on the proposed rate expression (eq. (8)), this reaction appears first order due to the dominance of

vacant sites on the catalyst. Dominance of vacant sites would simplify the rate equation to $r_{CH4} = kp_{CO_2}^{1/2}p_{H_2}^{1/2}$ (where p is the partial pressure of each species), indicating an overall reaction order of one on a Ni/Al catalyst.

MATLAB was also utilized to determine the reaction constant at the temperatures 250, 270, and 300 °C. by fixing the parameter $p = k_a\tau$, which was optimized to have minimal error. This optimization is done for each reaction order, although the only reaction order relevant to this study is first order. Once p was optimized, it was converted back into reaction constant and found they were found to be k as 0.301±0.012, 0.4741±0.009, 0.6293±0.019 s^{-1} for 250 , 270 , and 300 °C respectively.

Apparent Activation Energy & Frequency Factor

Following equation (6), values of ln(k) vs $\frac{1}{T}$ were plotted:

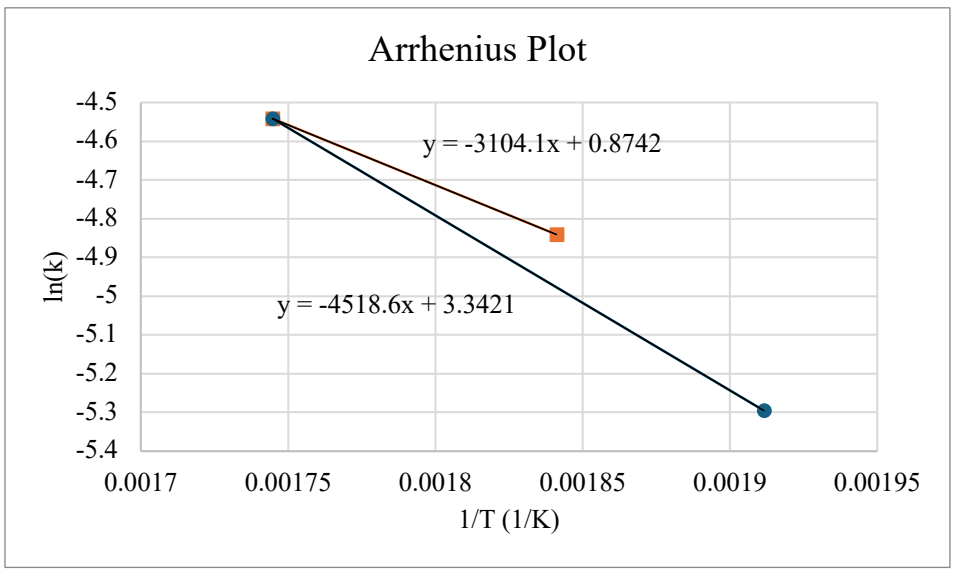

Figure 5: Arrhenius Plot Comparing 250 & 270 °C to 300 °C

Utilizing both the slope and the y-intercept, it was found that:

Table 2: Apparent Activation Energy & Frequency Factor at Varying Temperatures

	Ea (kJ/mol)	A (s^{-1})
250 vs 300 °C	25.81	28.28
270 vs 300 °C	37.57	2.40

The change in the slope indicates a change in the apparent activation energy of the reaction. The nonlinearity of the data is a potential indicator that the reaction is shifting from being limited by activation energy of the reaction towards being limited by the diffusion of the reactants to the catalytic sites.

Influence of Contact Time on Conversion

Finding the influence of contact time comes from varying the total flow rate and keeping the other parameters constant. The flow rates are as shown in Table 6 and were run at 270 ± 3 °C.

Contact Time:

Table 3: Varied Total Flow rate at Isothermal Conditions (T = 275 °C)*

N_2 (SCCM)	CO_2 (SCCM)	H_2 (SCCM)	Total (SCCM)	Theoretical CO_2 inlet (Vol %)	Actual CO_2 inlet (Vol %)	IR reading (Vol %)	Conversion
180	200	180	560	0.0143	0.0183	0.0119	0.3514
225	250	225	700	0.0143	0.0174	0.0119	0.3159
270	300	270	840	0.0143	0.0168	0.0120	0.2863
315	350	315	980	0.0143	0.0165	0.0122	0.2650
360	400	360	1120	0.0143	0.0162	0.0124	0.2358

*Note: There is an error within T of ± 3 °C. This is because the apparatus' PID failed to stabilize and fluctuated between $272\ °C < T < 278$ °C.

Since we are varying in total flowrate, we must compute residence times at each q, $\tau = \frac{V}{q}$. The results are summarized in Table 9.

The results showed that increasing contact time by decreasing the total flow rate increases the overall conversion. Since we are varying in total flowrate, we must compute residence times at each q, $\tau = \frac{V}{q}$.

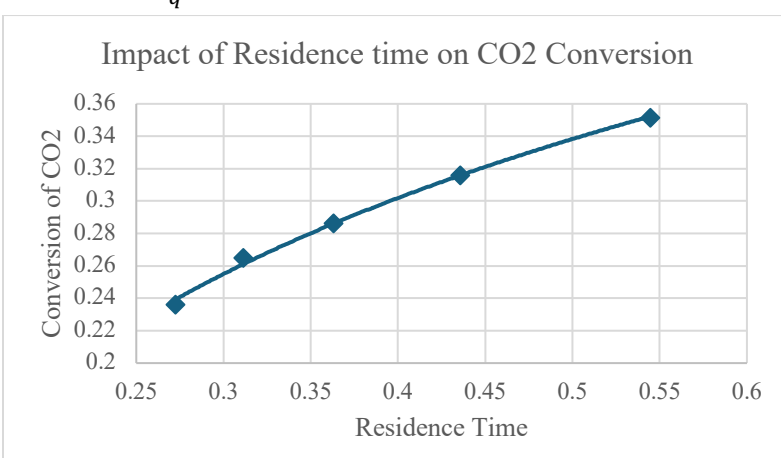

Figure 6: Impact of Residence Time on CO_2 Conversion

Conclusions

The use of CO_2 as the feed for methanation could potentially be a viable operation for reducing greenhouse gas emissions while establishing a renewable way to generate synthetic natural gases. This set of experiments investigated reaction parameters in a CSTR utilizing a 10 wt.% Ni/Al_3O_2 under stoichiometric conditions. The reaction was determined to be a first order reaction, with the reaction constants found to be 0.301 ± 0.012, 0.4741 ± 0.009, 0.6293 ± 0.019 s^{-1} at 250, 270, and 300 °C respectively. Additionally, the Ea for the reaction was found to be 25.81 and 37.568 kJ/mol with frequency factors 28.28 and 2.40 s^{-1}. Future work on this study should be to determine whether the reaction has reached its diffusion limitation by testing at higher temperatures and their impact on the apparent activation energy.

Notation

Term	Symbol	Units
Temperature	T	°C or K
Flowrate	q	SCCM
Concentration	C_A	Vol %
Rate constant	k	s^{-1}
Activation energy	Ea	J
Frequency Factor	A	s^{-1}
Gas Constant	R	$\frac{J}{K \cdot mol}$
Volume	V	mL
Pressure	P	Pa
Dimensionless Correlation Number	M_{an}	-
Rate	r	$\frac{mol}{s}$
Residence time	τ	s or min
Overall kinetic order	Θ	-

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